

EDA CAPSTONE PROJECT: PROBLEM STATEMENT

Indian E-Commerce Transaction Analysis (2M Records | 2GB Dataset)

Course: PG-DBDA - Statistics & Data Insights Module

Institution: C-DAC Kharghar, Navi Mumbai

Dataset Size: ~2,000,000 records (300MB CSV, expands to 2GB in memory)

Date Range: January 2022 – December 2024 (3 years)

1. BUSINESS CONTEXT

Scenario

You work as a **Data Analytics Lead** for a rapidly growing Indian e-commerce platform (similar to Flipkart, Amazon, Myntra).

Your company operates across all major Indian states and handles millions of transactions annually.

Key Business Challenges:

- **Rising Return Rates:** Customer returns increased from 12% (2022) to 18% (2024)
- **Inconsistent Delivery Performance:** Some states have 95% on-time delivery; others only 60%
- **Payment Mode Fragmentation:** Over-reliance on COD (₹ on cash handling costs); low UPI adoption in tier-2 cities
- **Category-Specific Issues:** Electronics returns are 3x higher than Books; why?
- **Regional Revenue Gap:** Maharashtra & Delhi generate 60% of revenue; other states underperform

Your Mission

Perform a comprehensive Exploratory Data Analysis (EDA) on 2M messy transactions to:

1. Identify root causes of business problems
2. Uncover hidden patterns in customer behavior, regional performance, and product categories
3. Generate 5–10 **actionable, quantified recommendations** with ₹ impact

Success = Insights that drive business decisions & ₹ millions in impact

2. DATASET DESCRIPTION

File Location

ecommerce-transactions-raw.csv
Size: ~300 MB (2M rows × 30 columns)

Column Definitions & Data Quality Issues

#	Column Name	Data Type	Range/Values	Quality Issues	Cleanin g Priority
1	transaction_id	String (UUID)	TXN_0000000001	Duplicates (1.5%), NULLs (5%)	 HIGH
2	user_id	Integer	10,000–500,000	Negative IDs (~50), NULLs (2%)	 HIGH
3	product_id	String	PID_123456	Inconsistent format, NULLs (1%)	 MED
4	product_category	Categorica l	8 categories (Electronics, Clothing, Books, Home, Sports, Beauty, Groceries, Furniture)	Case mismatch (Electronics vs electronics), typos, NULLs (0.5%)	 MED
5	sub_category	Categorica l	5 per category	Same as category, case issues	 MED
6	product_price	Float	₹500–₹50,000	Skewed distribution (outliers >₹200K), NULLs (3%)	 HIGH

7	discount_percent	Integer	-10 to 120	Negatives (-10): 96K records (data error), >100%: 96K records (invalid), NULLs (2%)	 HIGH
8	final_price	Float	Calculated	Inconsistent with $\text{price} \times (1 - \text{discount}/100)$ in 100K records (~5% error rate)	 HIGH
9	quantity	Integer	1–1000	Zeros: 0, Negatives: 0, Extremes >100: rare outliers	 LOW
10	transaction_date	String/Date	2022-01-01 to 2024-12-31	Mixed formats (YYYY-MM-DD, DD/MM/YYYY, DD-MM-YYYY), NULLs (0.1%)	 HIGH
11	payment_mode	Categorical	COD, UPI, Card, Net Banking, Wallet, NULL	Case inconsistency (COD vs cod), NULLs (2%), Extra spaces	 MED
12	order_status	Categorical	Completed, Pending, Cancelled, Returned	Extra spaces, case mismatch, NULLs (1%)	 MED

13	delivery_date	String/Date	Same format issues	NULLs (20%: indicates undelivered orders), Mixed formats, Delivery before transaction: 40K records (logical error!)	 HIGH
14	delivery_days	Integer	3–15 (normal), Negatives (2%), NULLs (20%)	Negative values when delivery < transaction date, Inconsistent with calculated (delivery_date – transaction_date)	 HIGH
15	return_flag	Integer	0 (not returned), 1 (returned)	Inconsistent encoding (0/1 vs Yes/No), NULLs (1%)	 MED
16	return_reason	String	Defective, Wrong Item, Quality Issue, Size Mismatch, Changed Mind, Better Price Found	NULLs only for return_flag=0 (40K mismatch cases), Typos, Case issues	 HIGH
17	return_date	String/Date	Same format issues	NULLs only for return_flag=0 (40K mismatch), Mixed formats, return_date < transaction_date in rare cases	 HIGH

18	seller_rating	Float	1.0–5.0 (stars)	NULLs (10%: 200K), Clipped to [1, 5], Outliers (values like 0.0, 6.0 before clipping)	● MED
19	customer_rating	Float	1.0–5.0 (stars)	Same as seller_rating	● MED
20	buyer_state	Categorical	13 states (MH, DL, KA, TN, GJ, AP, UP, WB, HR, PN, RJ, MP, Others)	Abbreviations vs full names mixed (~5% of records), NULLs (1%), Extra spaces	● MED
21	seller_state	Categorical	Same as buyer_state	Same issues	● MED
22	customer_segment	Categorical	Premium, Standard, Basic	Inconsistent naming (premium vs Premium), NULLs (2%)	● MED
23	gst_amount	Float	Calculated (5%, 12%, 18% based on category)	NULLs (2%), Inconsistent calculation (should be $\text{final_price} \times \text{quantity} \times \text{rate}$), Negatives (rare, ~100)	● MED
24	shipping_cost	Float	₹0 (free for orders >₹500), ₹50–₹200 (below ₹500)	NULLs (1%: ~10K), Negatives (10K records: data entry errors), Illogical values >₹500 (~5K)	● HIGH

25	refund_amount	Float	Calculated (for return_flag=1 only)	NULLs for non-returned items is OK; Mismatches with final_price × quantity (~3%), Negatives (rare)	● MED
26	device_type	Categorical	Mobile, Desktop, Tablet	Case mismatch, NULLs (0.5%)	● LOW
27	browser_type	Categorical	Chrome, Firefox, Safari, Edge, Opera	Same case issues, NULLs (1%)	● LOW
28	session_duration_min	Float	5–500 minutes (normal), Extremes up to 1200 (outliers)	Extreme outliers (0.1% of records: >600 min = possible data corruption), NULLs (0.5%)	● MED
29	has_reviews	Integer	0 (no), 1 (yes)	NULLs (1%), Inconsistent encoding	● LOW
30	loyalty_member	Categorical	Yes, No, NULL	Inconsistent (Yes vs yes), NULLs (5%: 150K)	● LOW

3. SPECIFIC BUSINESS QUESTIONS TO ANSWER

A. RETURN ANALYSIS (Root Cause: Why 18% Return Rate?)

1. **Which product categories have the highest return rates?** (Expected: Electronics > Books > Clothing)
2. **What are the top return reasons by category?** (Is it defective items, size mismatch, or buyer's remorse?)
3. **Do return rates differ significantly by state?** (Urban vs rural states)
4. **Correlation between seller_rating and return likelihood?** (Do low-rated sellers have higher returns?)

5. **Days to return: Is 30 days an optimal window?** (When do 80% of returns happen?)

B. DELIVERY PERFORMANCE (Issue: 60–95% on-time delivery variance)

1. **On-time delivery % by state** (Target: $\geq 90\%$ for all states)
2. **Average delivery days by category** (Which categories take longest?)
3. **Impact of distance (buyer_state vs seller_state) on delivery time**
4. **Seasonal patterns:** Do monsoon months (June–Sept) have worse delivery?
5. **Is delivery performance affecting customer ratings?** (Correlation)

C. PAYMENT MODE ANALYSIS (Challenge: COD costs ₹50–100 per transaction)

1. **Payment mode distribution:** COD, UPI, Card, Net Banking, Wallet (%)
2. **Payment mode adoption by state:** Which states prefer UPI? Which rely on COD?
3. **Payment mode by customer_segment:** Do Premium customers prefer Card/UPI?
4. **Payment mode by device_type:** Mobile-first users = UPI? Desktop = Card?
5. **Return rate by payment mode:** Does COD have higher returns? (Trust issue?)
6. **₹ Impact:** If we shift 20% of COD to UPI, save ₹X per year?

D. REGIONAL PERFORMANCE (52% of revenue from 2 states—concentration risk)

1. **Revenue by state** (Total: $\text{final_price} \times \text{quantity} \times 1.18$)
2. **Average Order Value (AOV) by state**
3. **Customer count by state** (market penetration)
4. **Return rate by state** (Is MH performing better than UP?)
5. **Top performing state: What makes it successful?** (Higher ratings, faster delivery, better category mix?)
6. **Opportunity states:** Where can we expand?

E. CATEGORY PERFORMANCE

1. **Revenue contribution by category** (ranking)
2. **Profit margin by category** ($\text{revenue} - \text{cost_of_goods} - \text{shipping} - \text{GST}$)
3. **Return rate by category** (Electronics = 25%? Books = 5%?)
4. **Customer rating by category** (Do high-quality categories get better ratings?)
5. **Average shipping cost by category** (Heavy items like Furniture cost more)
6. **Seasonal trends by category** (Electronics spike in Nov-Dec, Fashion in summer)

F. CUSTOMER SEGMENTATION & LTV

1. **Average Order Value (AOV) by segment** (Premium vs Standard vs Basic)
2. **Return rate by segment** (Do Premium customers return less?)

3. **Repeat purchase behavior** (How many unique orders per user_id?)
4. **Customer Lifetime Value (CLV) by segment** (Total spend per user over 3 years)
5. **Loyalty program effectiveness** (Do loyalty_member=Yes have higher repeat purchases?)
6. **Device preference by segment** (Do Premium customers use Desktop more?)

G. TIME SERIES & SEASONALITY

1. **Monthly revenue trend** (2022 vs 2023 vs 2024)
2. **Seasonal peaks** (Which months are best? Diwali, summer vacations, etc.)
3. **Day-of-week effect** (Weekends vs weekdays)
4. **Year-over-year growth rate** (Is the business growing?)
5. **Return seasonality** (Do winter clothes have winter-specific returns?)

H. OUTLIERS & DATA ANOMALIES (For data quality insights)

1. **Products with extreme prices** (₹50K+): Are they real? (Furniture, Electronics outliers)
 2. **Extreme delivery delays** (>30 days): How many? Why?
 3. **Extreme session durations** (>5 hours): Are they real or bot traffic?
 4. **Orders with zero GST**: Which categories? Is it a bug?
 5. **Negative shipping costs**: Is it refunds misclassified?
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4. DATA QUALITY CLEANING CHECKLIST

Phase 1: Deduplication & Null Handling

- Remove exact duplicates on transaction_id
- Handle missing values in delivery_date (20%): Are these undelivered orders? Keep or drop?
- Handle missing values in ratings (10%): Impute with median per category or category + state?
- Handle missing values in payment_mode (2%): Impute with mode or drop?

Phase 2: Type Casting & Format Standardization

- Parse all date columns to datetime (handle mixed formats: YYYY-MM-DD, DD/MM/YYYY, DD-MM-YYYY)
- Convert boolean columns (return_flag, has_reviews, loyalty_member) to consistent 0/1 or Yes/No
- Standardize categorical columns: UPPERCASE or Title Case?

- Trim whitespace from all string columns

Phase 3: Logical Consistency Fixes

- **Delivery before order:** Remove or flag 40K records where `delivery_date < transaction_date`
- **Price/discount inconsistency:** Recalculate `final_price` for 100K mismatches; document if we trust original or recalculated
- **Negative discounts & discounts >100%:** Mark as errors; either drop or impute with category median
- **Return reason without return flag:** Fix 40K cases where `return_reason != NULL` but `return_flag = 0`
- **Return date without return flag:** Similar fix
- **Delivery days mismatch:** Recalculate from `(delivery_date - transaction_date)`

Phase 4: Outlier & Error Handling

- **Negative shipping costs:** Are these refunds? Investigate; either drop or flag
- **Negative discounts (-10%):** Are these surcharges? Document decision
- **Negative prices/final_price:** Drop (0 occurrences expected, but check)
- **Extreme session durations (>600 min):** Are these bot sessions? Flag for fraud team or drop
- **Zero GST amounts:** Which categories? Is it a logic bug (5% categories = Books/Clothing)?
- **Extreme product prices (>₹500K):** Verify against category (Furniture outliers are OK; Clothing is not)

Phase 5: Validation & Documentation

- Log all cleaning decisions: "Removed 40K rows where `delivery_date < transaction_date` because..."
- Generate before/after comparison (original rows, null counts, outlier counts)
- Create a **Data Quality Report** (2–3 pages) documenting every decision

DELIVERABLES CHECKLIST

Code & Analysis

- **Jupyter Notebook** (capstone-eda.ipynb): Full pipeline with markdown narrative between code cells

- **Python script** (data_cleaning.py): Modular, reusable code for cleaning & transformation
- **Cleaned dataset** (ecommerce-transactions-cleaned.csv): Exported for reproducibility

Documentation

- **Data Cleaning Report** (2–3 pages, PDF): Every decision documented with rationale
- **Exploratory Data Analysis Report** (5–7 pages, PDF): Key findings, visualizations, insights
- **Executive Summary** (1 page): Top 3 recommendations with ₹ impact

Visualizations

Code Quality

- Code is clean, modular, well-commented
- Reproducible (all seeds set, data paths documented)
- Error handling for edge cases

Repository

- GitHub repo with README.md
- Folder structure: /data, /notebooks, /reports, /visualizations, /scripts
- .gitignore (exclude raw data, large CSVs)